GAMUDA HACKATHON · CHALLENGE 2: AI FOR EDUCATION & LEARNING

CikguAI: every student at their own pace, every teacher sees who needs help.

A bilingual (Bahasa Melayu + English) Socratic AI tutor, adaptive practice engine and teacher's assistant for Malaysian secondary schools, starting with KSSM Form 2 Mathematics and Science.

challenge2.rohaizat.cloud · one-click "Try as Student" / "Try as Teacher"

The problem

41%
of Malaysian 15-year-olds reached baseline maths proficiency in PISA 2022 (OECD average 69%).

397
Malaysia's PISA 2025 maths score, down another 11 points.

1 : 30
One teacher, thirty students, thirty different gaps, and marking and planning that eat the time to close them.

For students (mobile-first)

- **Socratic AI tutor:** hints and questions, not answers. Snap a photo of your working and it finds the exact wrong step.
- **Adaptive quizzes:** each question targets your weakest skill at ~70% predicted success.
- **"Your next 3 things to study"**, with reasons (weak, recent mistakes, unlocks the next topic).
- **XP, streaks, badges, daily goal** and a weekly class board that rewards effort.

For teachers

- **AI-assisted marking:** rubric-based draft grades with evidence, feedback and confidence. Nothing is released until the teacher approves.
- **Class insights:** mastery heatmap, at-risk students, repeated misconceptions, suggested groups, AI weekly summary.
- **Lesson planner:** KSSM-aligned plans built around the class's own data; edit, export to Word/PDF.
- **Worksheets:** bilingual questions with answer keys; approve the best into the adaptive bank.

The demo loop: from one wrong answer to a targeted lesson

1 Aiman gets a factorisation question wrong (sign error).

2 The tutor diagnoses the gap, Socratically.

3 Adaptive quiz targets it at the right difficulty.

4 Mastery rises live: 12% → 39% → 73%.

5 Teacher sees the 8-student sign-error cluster.

6 One click: a remedial lesson plan for that class.

How AI and ML are used

- ML **Bayesian Knowledge Tracing** estimates mastery of 13 KSSM skills after every answer.
- ML **Elo/Rasch ratings** calibrate question difficulty from real answers and pick the next question.
- Data Wrong options are tagged with the **misconception** they reveal; the dashboard counts repeated ones.
- Claude Tutoring (streamed), photo diagnosis, rubric grading, class summaries, lesson plans and question generation, all with schema-validated output.

Responsible by design

- Teachers approve every grade and every AI-written question before students see it.
- Mastery never depends on the AI; scores are clamped and totalled in code.
- Students are pseudonymised before class data is sent to the AI; photos are stripped of GPS and stored privately.
- Teachers see tutor use per topic, not the chats. Methodology and limitations are public.

What's next

Forms 1–5 and SPM subjects (Add Maths, Physics, Chemistry, Biology) · voice tutoring in BM and English · parent portal · offline-first PWA for rural schools · integration with school systems (e.g. Google Classroom) · fit BKT parameters per skill from pilot data and measure learning gains.

Stack: React · Fastify on Node 22 · SQLite · Claude (claude-opus-5-5) · KaTeX · nginx + Let's Encrypt. Sources: OECD PISA 2022 Country Note Malaysia; The Star 8 Sep 2026 (PISA 2025); Corbett & Anderson 1995; Pelánek 2016. Details at challenge2.rohaizat.cloud/methodology.